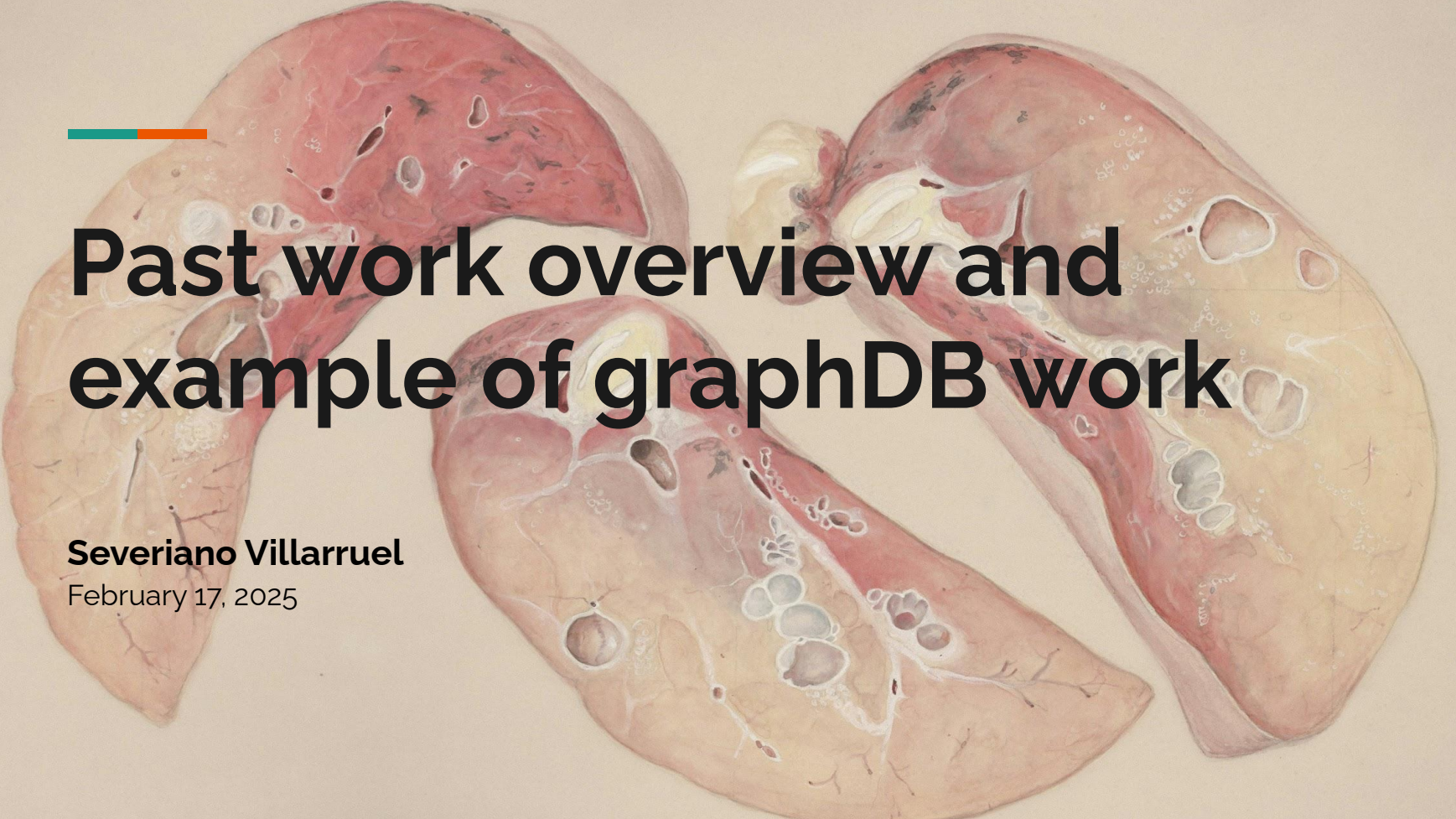




Past work overview and example of graphDB work

Severiano Villarruel

February 17, 2025

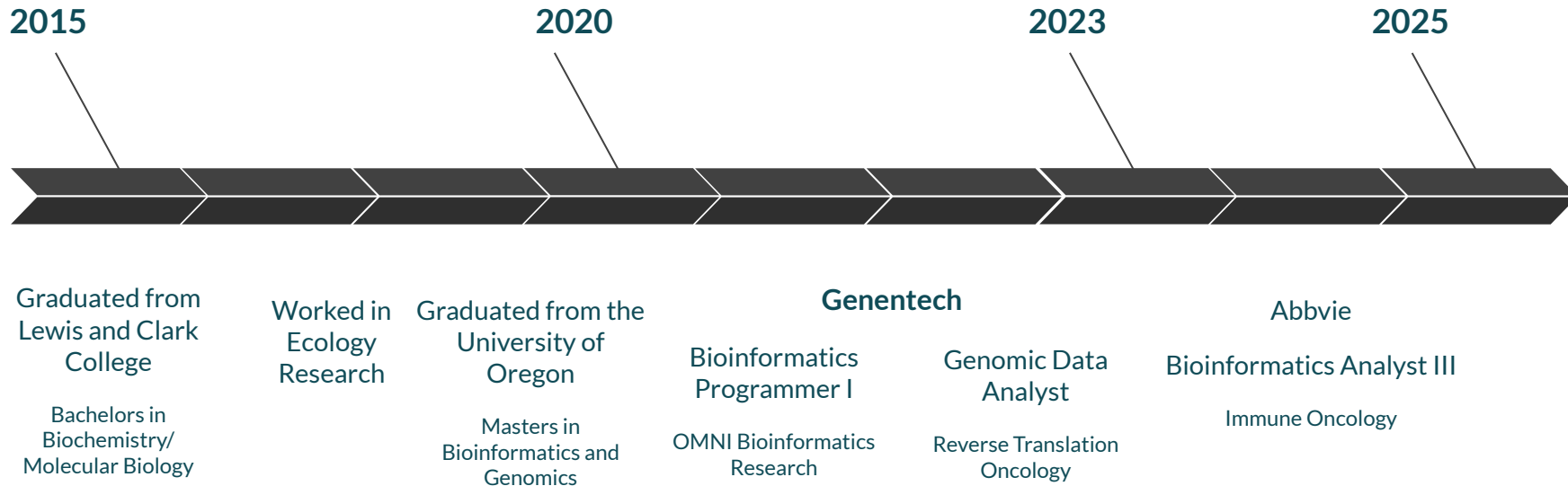


Intro

- Personal Background - 2 min
- Vignettes
 - Finding a genomic instability gene signature in real-world data - 10 min
 - Understanding the effects of target inhibition - 10 min
 - A semi-supervised method for cell type labeling - 10 min
 - Making a modern web app with WebGL - 3 min
- Improving Target Discovery Using a Graph Database
 - Novel Methods at Tempus - 5 min



Timeline



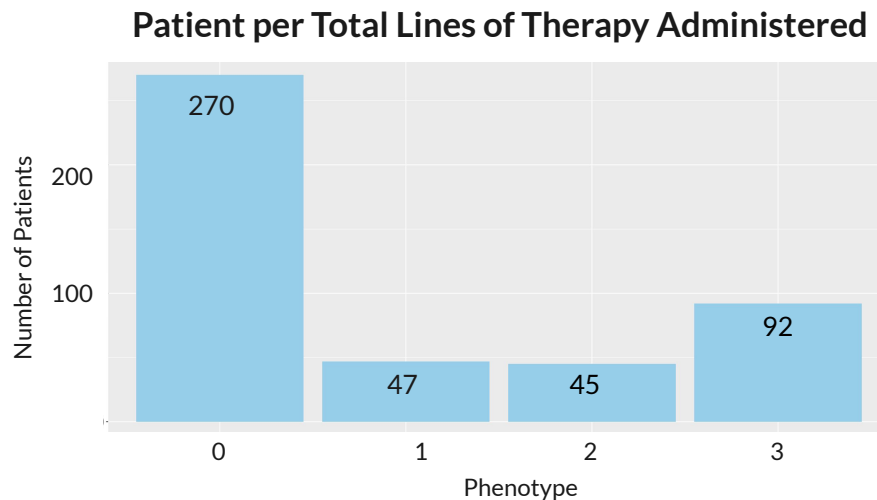


Finding a genomic instability gene signature in real-world data

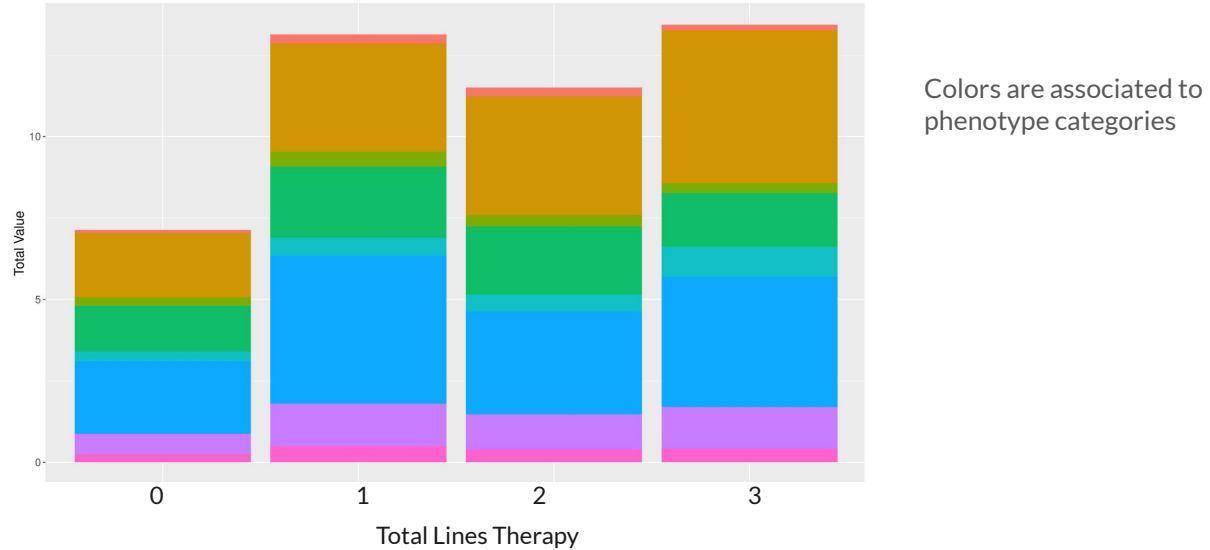
Genomic instability

Dataset Overview

- Real world blood cancer dataset
 - >400 patients
 - Age
 - Race
 - Cancer subtype
 - Lines of therapy
- Wrangle Data
- Transposable element calling
 - >2M elements across all patients
- Scaled and log normalized counts

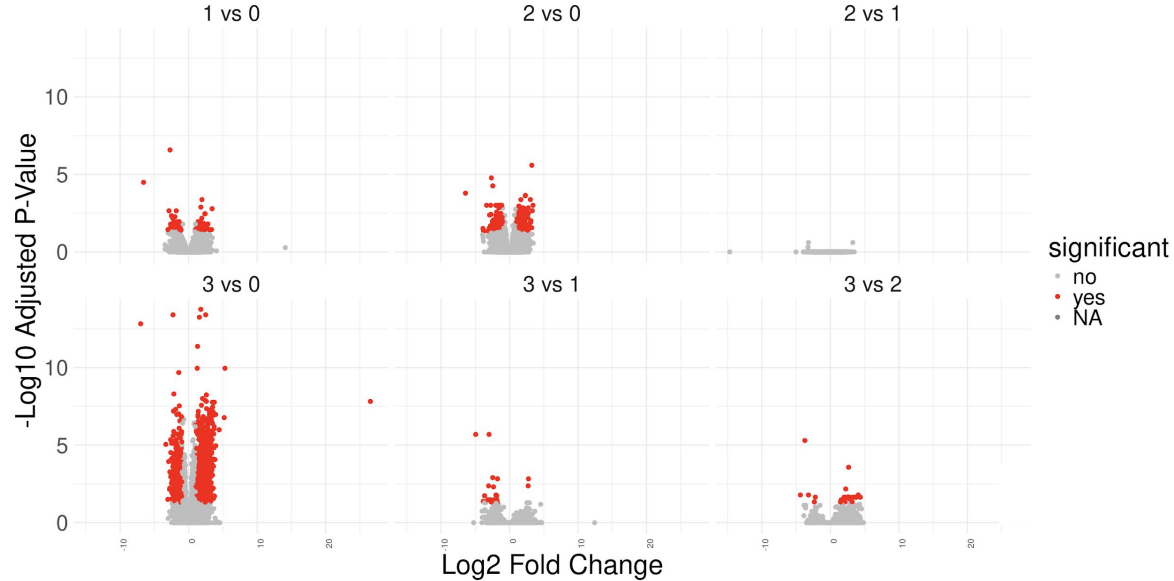


Distribution of TE families in each TLT



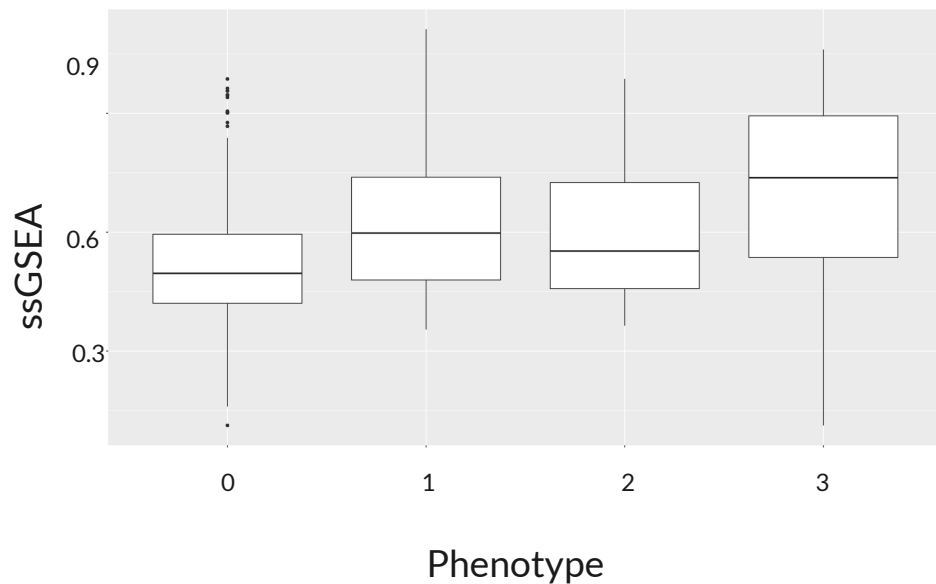
- The set of TEs in each TLT group are comprised of TEs from different families
- TE families are similarly distributed in each total lines of therapy

Transposable Element Differential Expression



- Pairwise differentially expressed TEs ~ TLT
- Expression change is present but varies in strength in each pairwise model
- Expression differences are greatest in TLT3 v TLT0

TE expression set marking instability

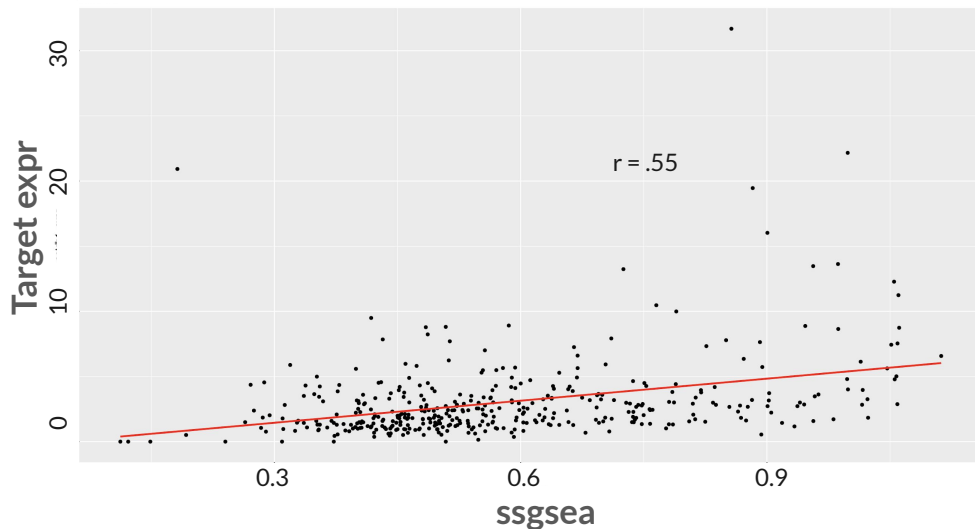


Top 10 DE TEs in TLT3 v TLT 0

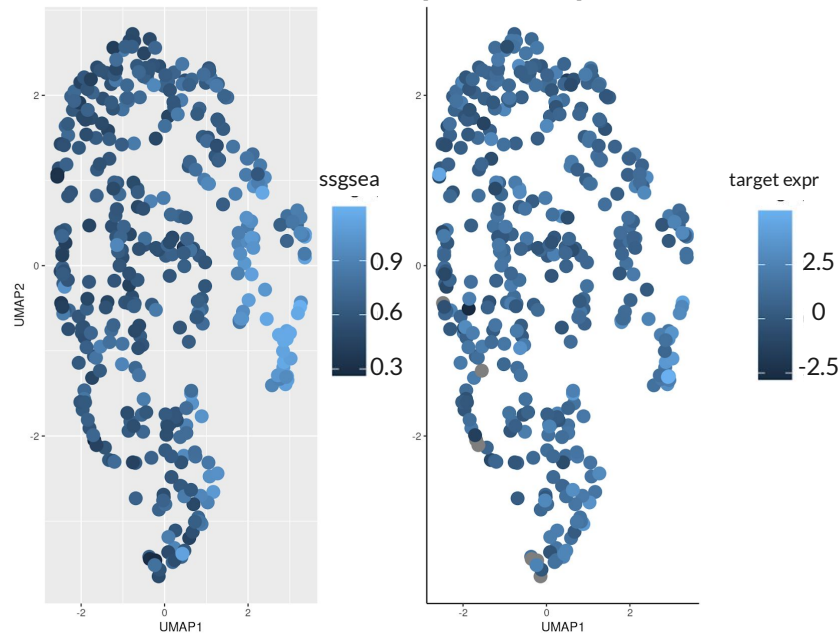
te_name	baseMean	log2FoldChange	lfcSE	pvalue	padj	repClass
TE-Q3	17.14092133	1.249369043	0.158898459	3.76E-15	1.10E-10	LINE
TE-W1	24.01318458	1.297445305	0.156426596	1.09E-16	4.26E-12	LINE
TE-E4	37.78200492	1.401946758	0.204631169	7.33E-12	6.60E-08	LINE
TE-R1	41.76270196	1.570215015	0.177641716	9.64E-19	5.64E-14	LINE
TE-T5	20.66389245	1.767351772	0.193754796	7.40E-20	1.73E-14	LTR
TE-Y8	18.35593535	1.864902715	0.265252918	2.06E-12	2.67E-08	LTR
TE-A9	13.14641258	2.030195638	0.281044718	5.06E-13	9.86E-09	LINE
TE-S2	2.415091121	2.408146909	0.335546374	7.14E-13	1.28E-08	LINE
TE-D6	5.580348939	2.490555837	0.279082088	4.49E-19	3.95E-14	LINE
TE-F5	2.155477025	2.580879713	0.361725025	9.68E-13	1.51E-08	LINE

Target expression and TLT signature score

Correlation of signature score and target expression



UMAP of TE expression space

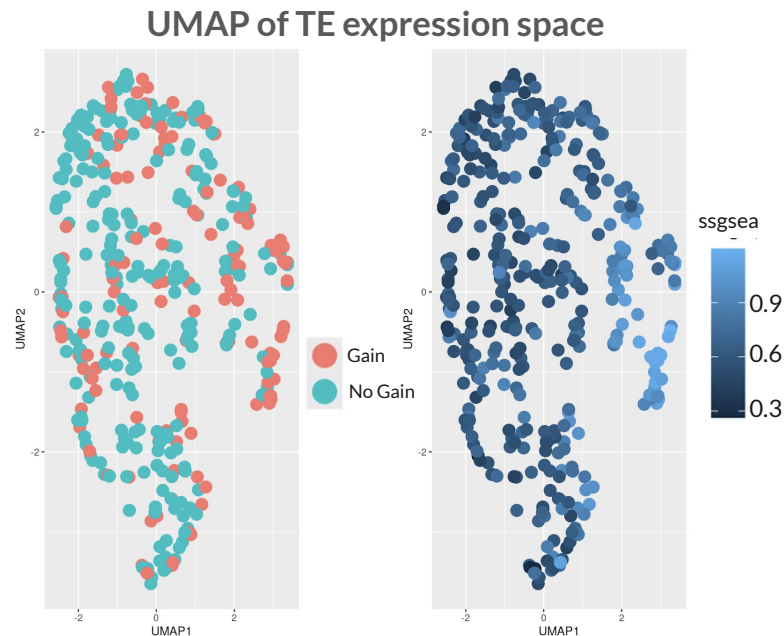


- Target expression appears to be correlated with genomic instability

Marker set composed of top 25 DE TEs in TLT3 v TLT0

Genomic Instability

1q Gain and TLT signature score



- ssGSEA score is significantly higher (t.test $p=2.86e-11$) in samples with a 1g Gain

Marker set composed of top 25 DE TEs in TLT3 v TLT0

Genomic Instability



Understanding the effects of target inhibition

Target inhibition

Data Summary



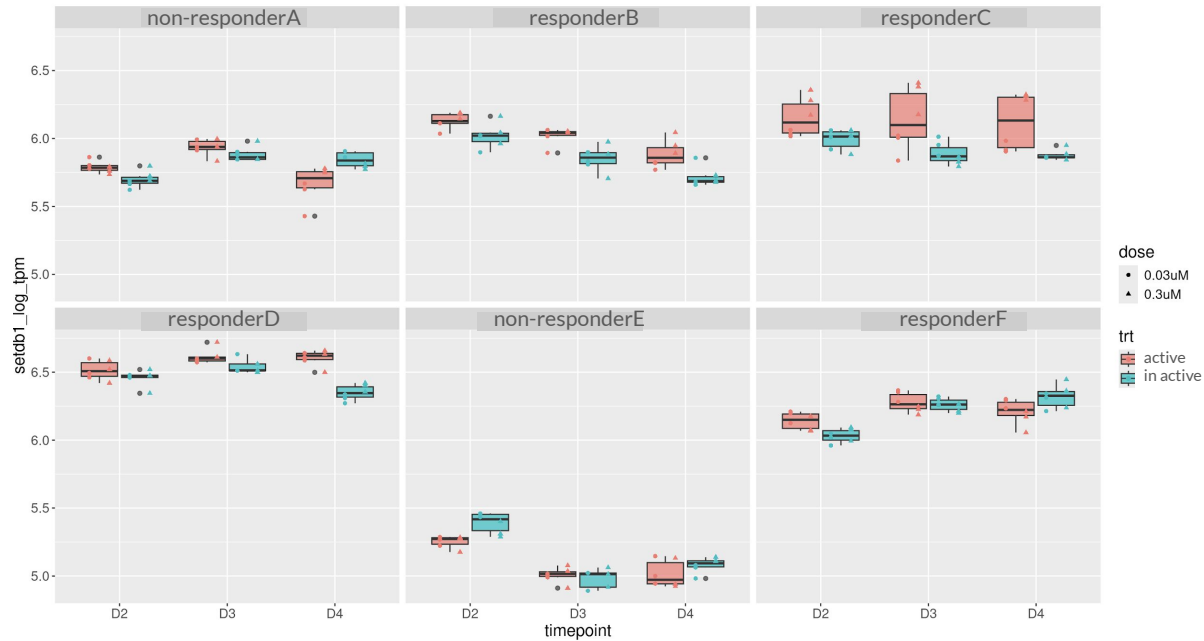
Overview

- RNA-seq from >200 samples
- 6 AML cell lines
 - Representing different stages of the cancer subtype

Phenotypes

- 2 doses
 - 0.3uM, 0.03uM
- 2 compounds
 - inactive
 - active
- 3 timepoints
 - D2, D3, D4

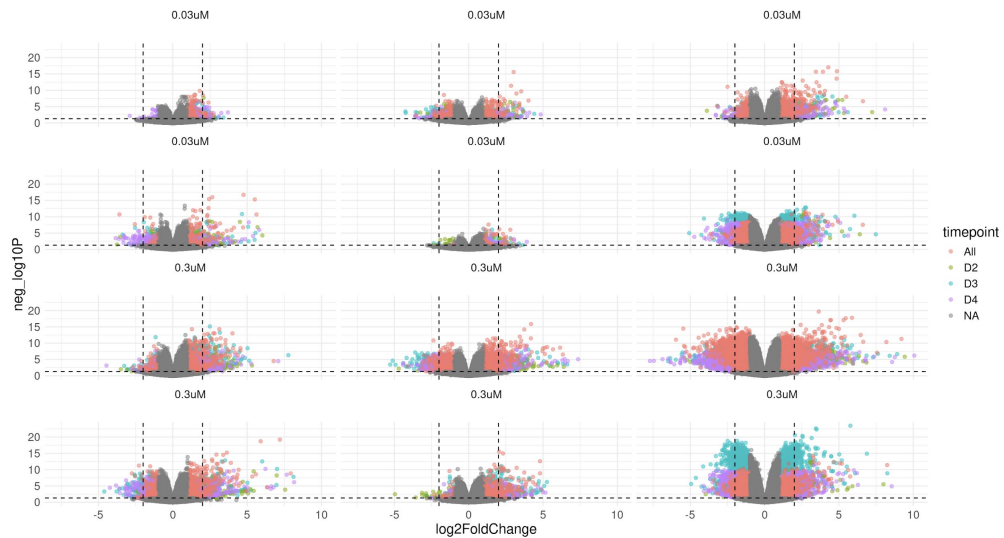
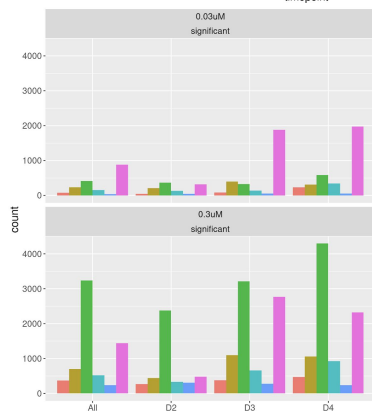
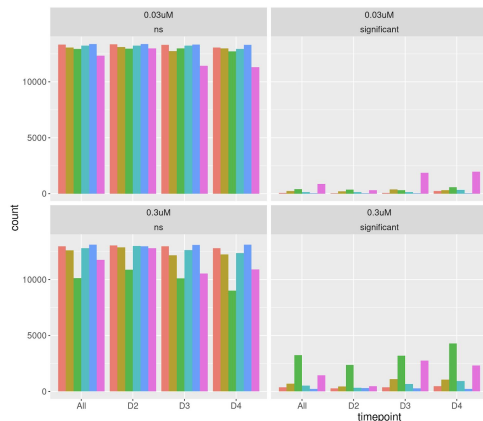
Target Expression by cell line



Target expression is highly variable across cell lines, and suggestively associated with response status

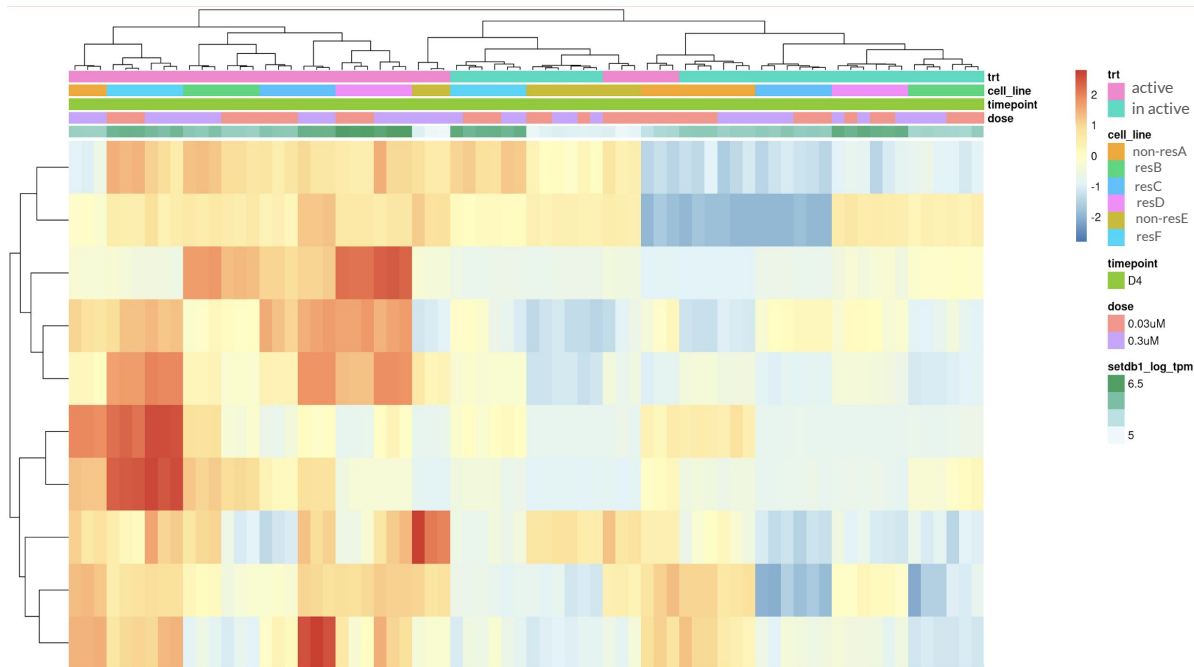
Target Inhibition

Quantification of differential expression across conditions



Target Inhibition

Cross cell-line differential expression results



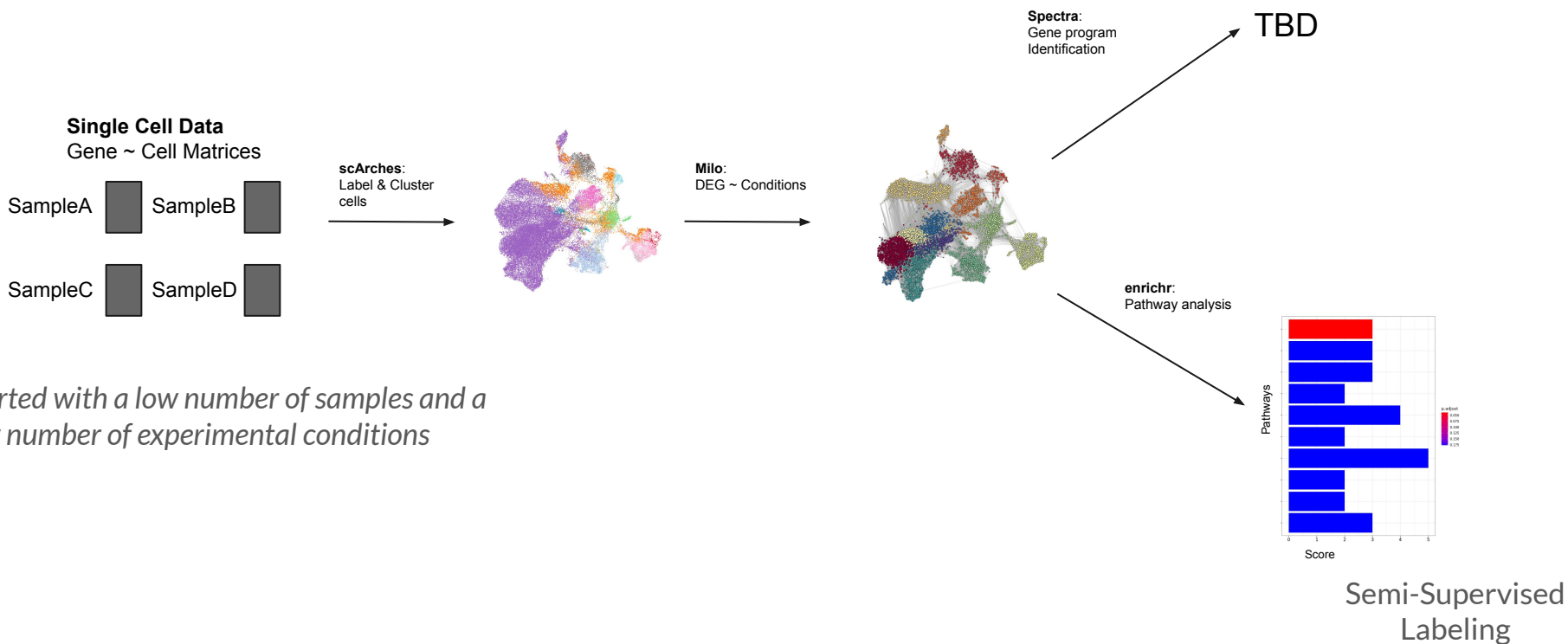
** Here at day 2 it can be seen that a gene set appears to be activated across the cell lines*



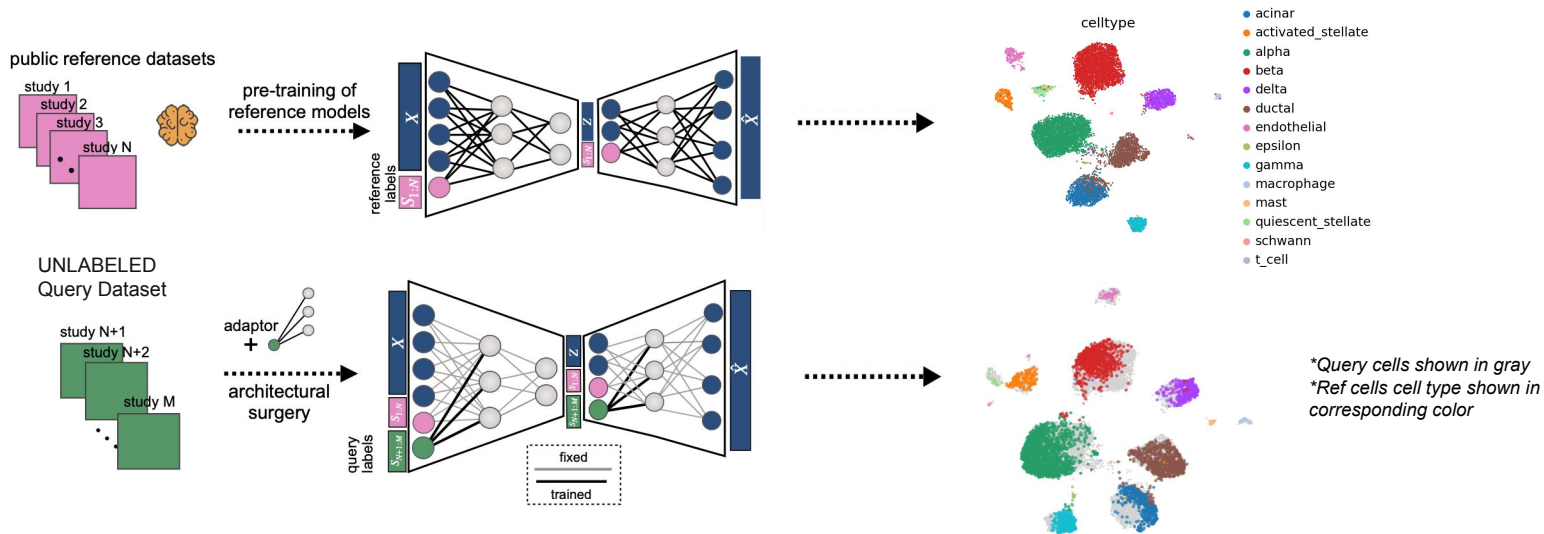
A semi-supervised method for cell type labeling

Using architectural surgery to label scRNA-seq data

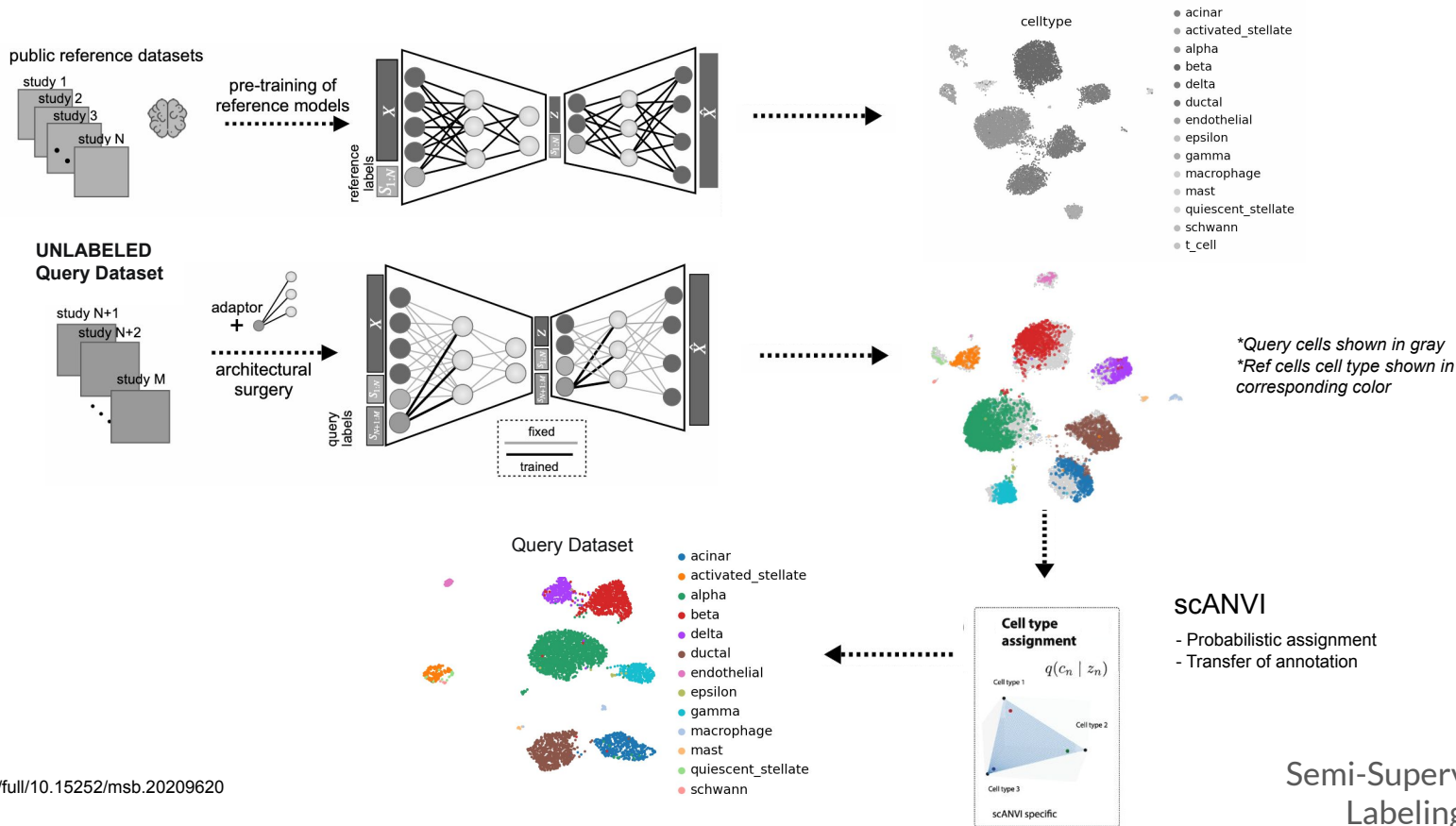
Processing + Analysis Overview



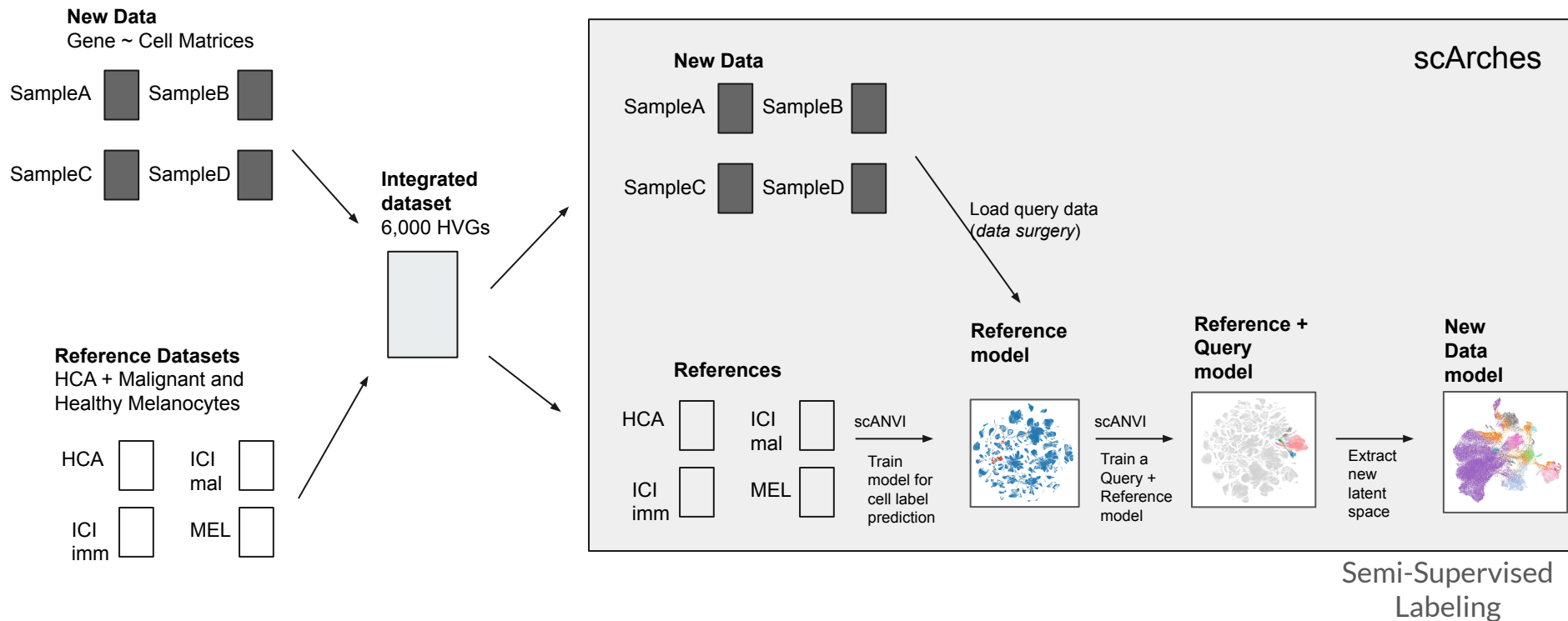
scArches: Map query cells to reference latent space



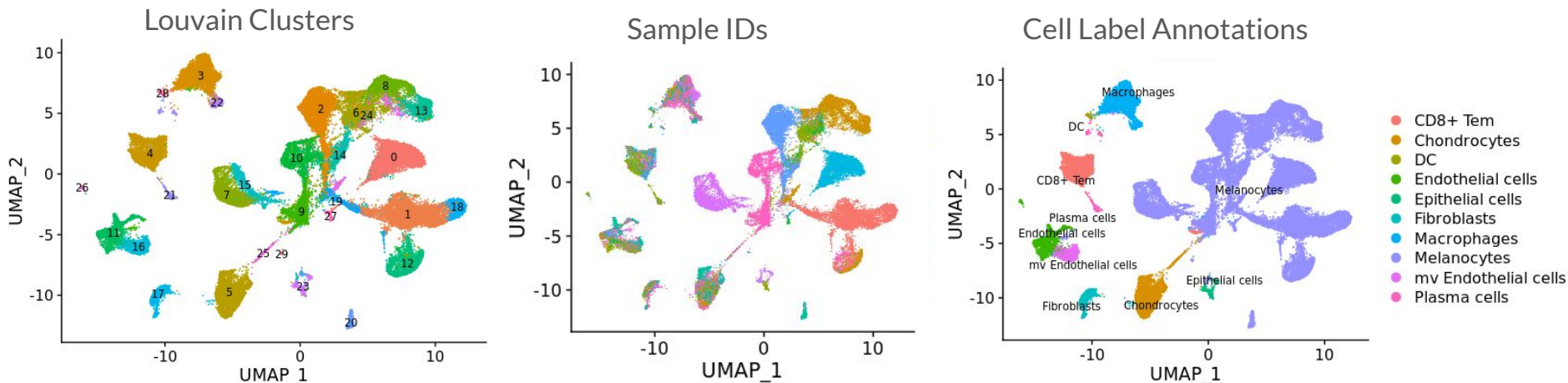
scArches: Map query cells to reference latent space



Label and Cluster Cells with scArches



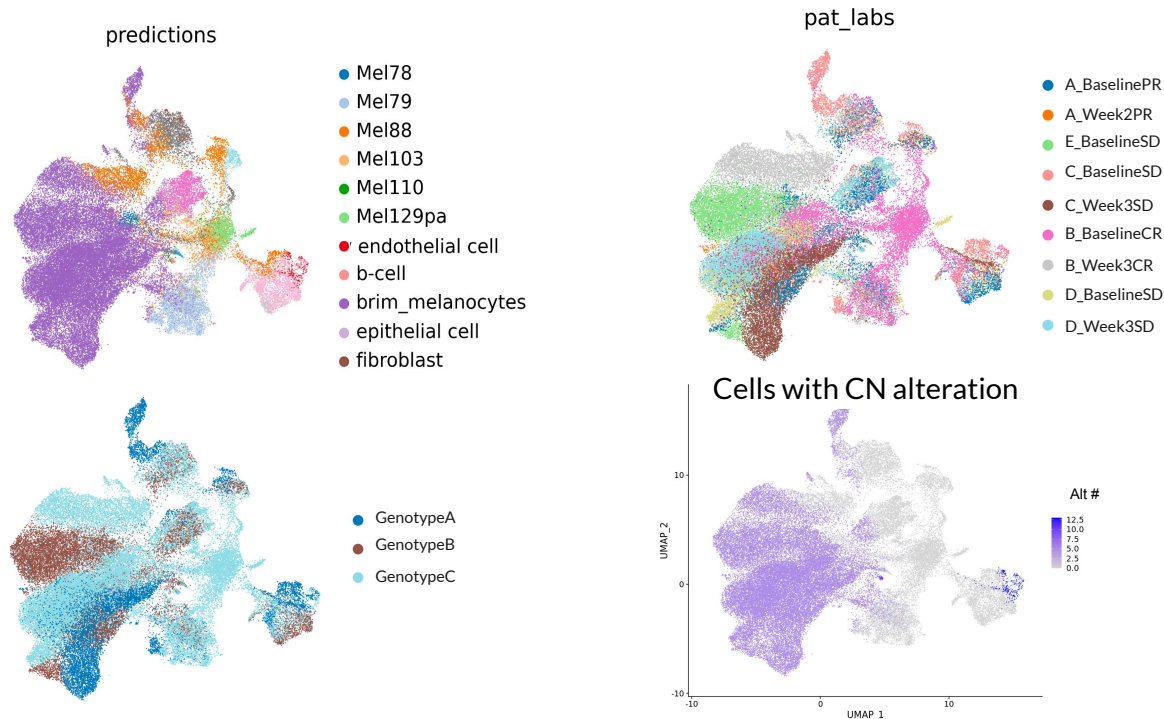
Correlation Based Cell Annotation



Cell annotations were added with BlueprintEncode reference using SingleR

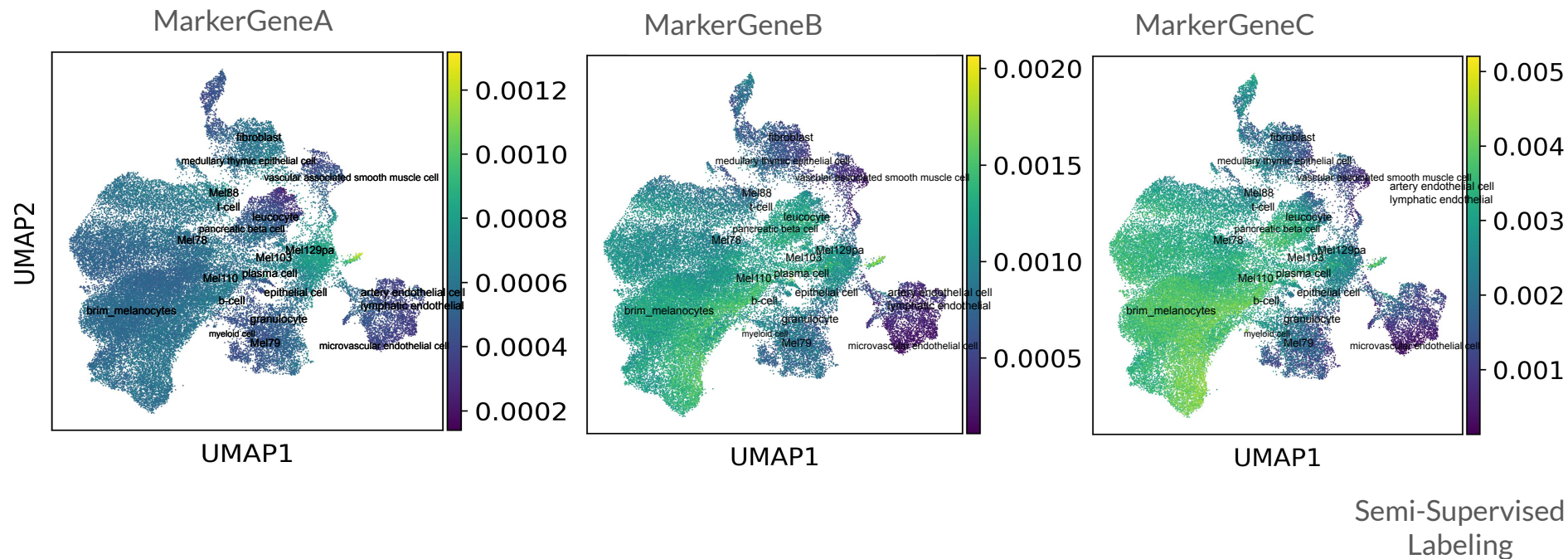
Semi-Supervised
Labeling

New model clustering metadata



Semi-Supervised
Labeling

Validating new model clustering using marker genes

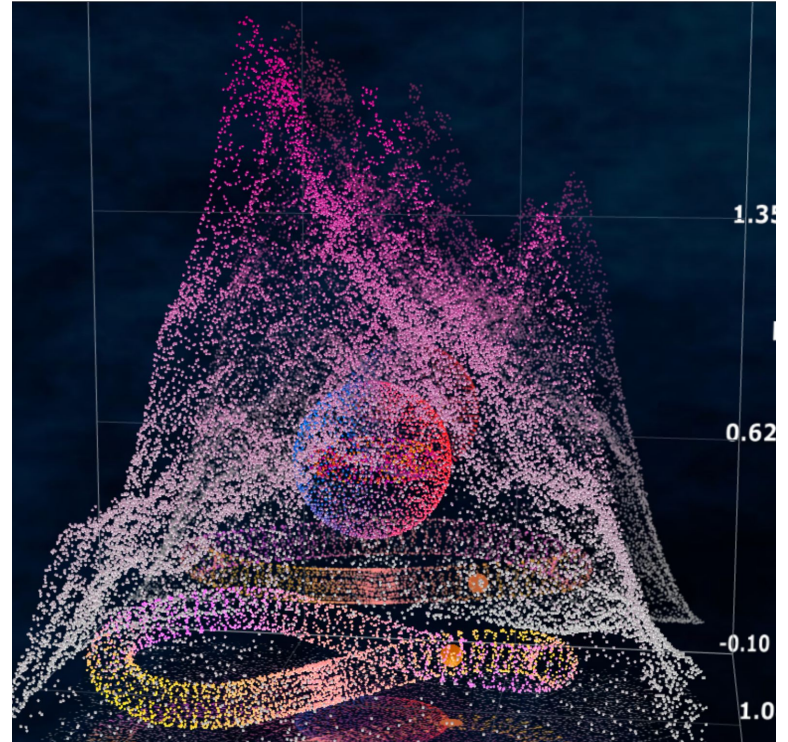




Breast Single Cell App

<https://pair-code.github.io/scatter-gl/>

<https://miabellaai.net/index.html>



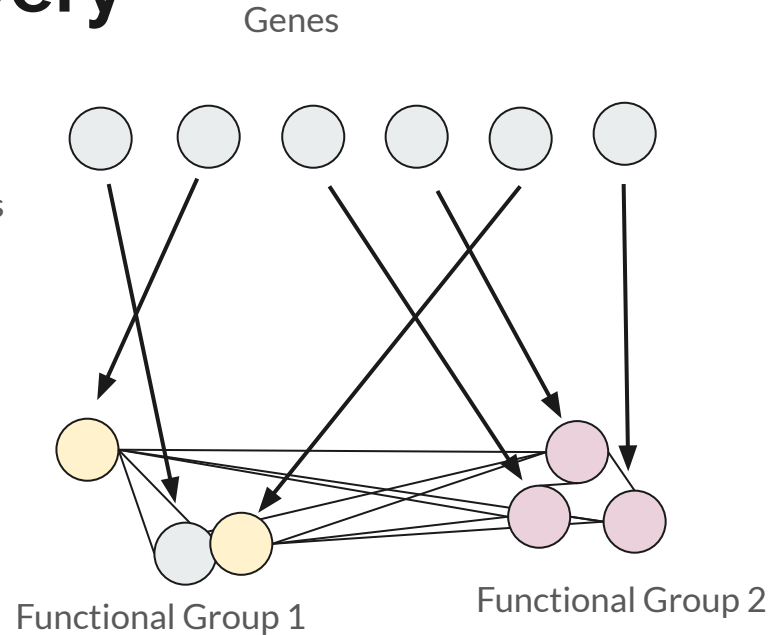


Improving Target Discovery Using a Graph Database

Novel Methods at Tempus

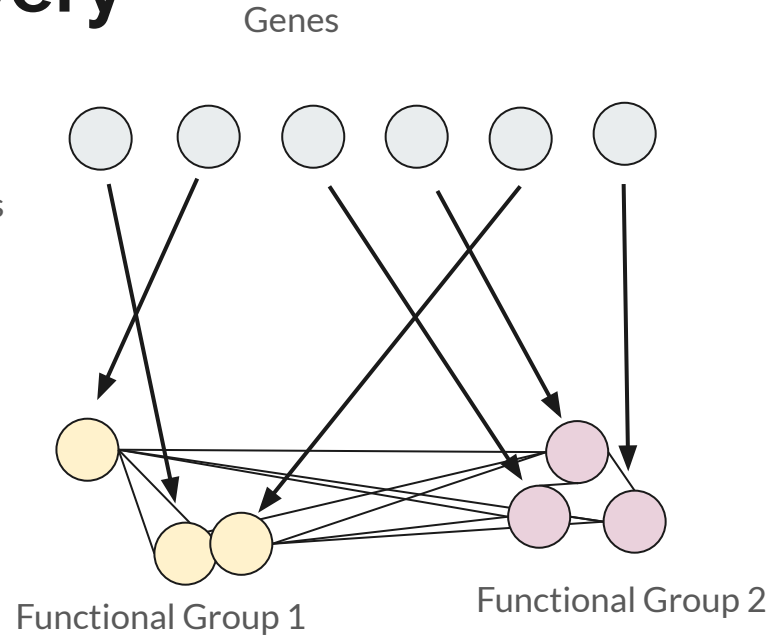
Improving Target Discovery

- Store gene data in a meaningful way
- Uncover biological function and sub-functions
- Combine knowledge from independent datasets
- Identify genes involved in cancer biology



Improving Target Discovery

- Store gene data in a meaningful way
- Uncover biological function and sub-functions
- Combine knowledge from independent datasets
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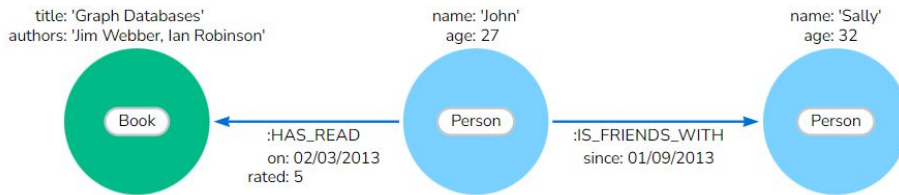
Utilizing CRISPR screens to identify functional groups

- CRISPR screens offer a powerful insight into the importance of a gene in a cell line
- DepMap Example
 - Dependency score

	Cancer Subtype	
	GeneA	GeneB
CellLine1	-1.2	-1.0
CellLine2	-0.8	-0.9
CellLine3	0	0
CellLine4	-0.3	-0.3

Graph Modeling with Neo4j

- Store data as nodes (entities) and relationships (edges)
 - Edges can have types and properties too
- Learn graph embedding to make predictions (missing edges between genes based on known connections)
- Easy access data with basic queries
- Built-in community detection algorithms
- NLP wrapper
 - Example:
 - User: "Which genes are essential in lung cancer?"
 - NLP converts → MATCH (g:Gene)-[:ESSENTIAL_IN]->(c:CancerType {name: "Lung"}) RETURN g
 - Neo4j retrieves relevant genes.



<https://neo4j.com/docs/getting-started/data-modeling/guide-data-modeling/>

Putting it all together

- Build a database that stores genetic data meaningfully
- Find new functional and sub-functional groups
- Combine knowledge across datasets
- Discover new cancer biology targets

